
Low-Resource Machine Translation: From Neural MT to LLMs

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Event: LU Talk

Date: 1 August 2026

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About My Research Journey

Master's Research — SIIT, Thammasat University

Low-resource neural machine translation for Myanmar, Thai, and English

Industrial Research — Brillar

One year developing generative AI and LLM-based applications

Current PhD Research — Institute of Science Tokyo

Low-resource English–Myanmar machine translation using large language models

Research Assistant — Institute of Science Tokyo

Olfactory language understanding and generation

Part 1: Overview of Low-Resource Machine Translation

What is Low-Resource Language?

A low-resource language has limited resources for developing and evaluating language technologies.

Resources:

- Parallel and monolingual data
- Annotated datasets
- Dictionaries and grammar resources
- Evaluation benchmarks
- Pretrained model coverage
- Computational and research support

Challenges for Low-Resource Language

Data

- Limited parallel and annotated corpora
- Domain and quality mismatch

Language characteristics

- Script and segmentation challenges
- Different morphology and word order

Evaluation

- Few reliable benchmarks
- Limited human evaluators

Model coverage

- Underrepresentation in pretraining
- Weak tokenization and multilingual transfer

What is Machine Translation?

Machine translation is the use of computational techniques to translate text or speech from one language to another, including the contextual, idiomatic, and pragmatic nuances of both languages. ([wiki](#))

Three main MT approaches:

- 1) Rule-based Machine Translation (RBMT)
- 2) Statistical Machine Translation (SMT)
- 3) Neural Machine Translation (NMT)

Statistical Machine Translation

There are four SMT models :

- Phrase Based Machine Translation (PBSMT)
- Hierarchical Phrase-based Statistical Machine Translation (HPBSMT)
- Operational Sequence Model (OSM)
- Syntax-based Machine Translation.

Neural Machine Translation

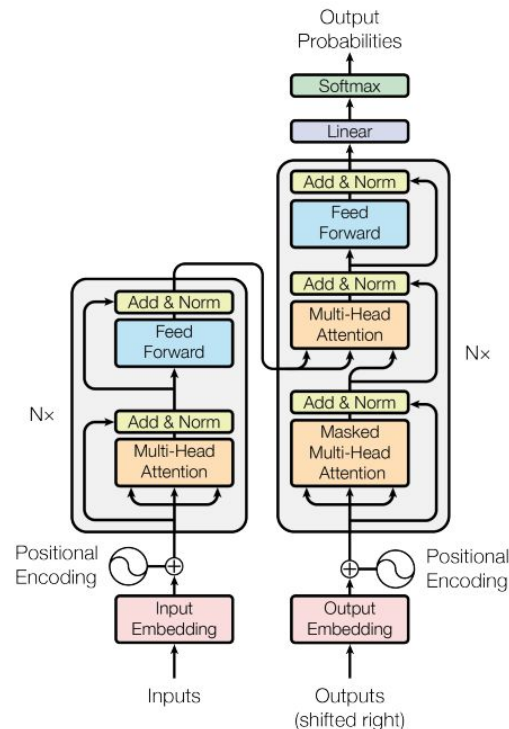
Types of NMT models:

- 1) Recurrent Neural Network (RNN) Based NMT
- 2) Convolutional Neural Network (CNN) Based NMT
- 3) Transformer
- 4) Pretrained Models

Transformer

Transformer has an attention-based architecture for sequence-to-sequence learning.

- It follows an encoder-decoder architecture.
- The encoder converts the input sequence into contextual representations.
- The decoder generates the output sequence one token at a time.
- The decoder operates autoregressively, using previously generated tokens to predict the next token.
- Due to attention mechanism, Transformers enable more efficient parallel training.
- Transformer has become the foundation of many modern NLP models, including neural machine translation systems and large language models.



Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

Pre-trained Model

Pretraining

- Learn general language patterns from large-scale text
- Use both labeled and unlabeled data

Adaptation

- Fine-tune the model for a specific downstream task
- Require much less task-specific training data
- Strong performance across many NLP tasks
- Useful for low-resource settings [1]

Examples

BERT [2] · mBART [3] · mT5 [4] · ELECTRA [5]

[1] Qiu, Xipeng, et al. "Pre-trained models for natural language processing: A survey." *Science China technological sciences* 63.10 (2020): 1872-1897.

[2] Kenton, Jacob Devlin Ming-Wei Chang, and Lee Kristina Toutanova. "Bert: Pre-training of deep bidirectional transformers for language understanding." *Proceedings of naacL-HLT*. Vol. 1. No. 2. 2019.

[3] Liu, Yinhan, et al. "Multilingual denoising pre-training for neural machine translation." *Transactions of the Association for Computational Linguistics* 8 (2020): 726-742.

[4] Xue, Linting, et al. "mT5: A massively multilingual pre-trained text-to-text transformer." *Proceedings of the 2021 conference of the North American chapter of the association for computational linguistics: Human language technologies*. 2021.

[5] Clark, Kevin, et al. "Electra: Pre-training text encoders as discriminators rather than generators." *arXiv preprint arXiv:2003.10555* (2020).

Large Language Models

- Transformer-based decoder-only generative models pretrained on large-scale text data.
- Learn broad linguistic and world knowledge through next-token prediction
- Perform multiple tasks through natural-language instructions
- Zero-shot and Few-shot learning
- Generate, summarize, translate, and reason over text
- Can adapt to new tasks using examples provided in the prompt

Challenges for low-resource languages

- Unequal representation in pre-training data
- Prompt sensitivity, limited reproducibility, hallucination
- Weak evaluation resources (Benchmarking)

Examples

GPT [1] · LLaMA[2] · Gemma[3] · Qwen[4]

[1] Achiam, Josh, et al. "Gpt-4 technical report." *arXiv preprint arXiv:2303.08774* (2023).

[2] Touvron, Hugo, et al. "Llama: Open and efficient foundation language models." *arXiv preprint arXiv:2302.13971* (2023).

[3] Team, Gemma, et al. "Gemma: Open models based on gemini research and technology." *arXiv preprint arXiv:2403.08295* (2024).

[4] Bai, Jinze, et al. "Qwen technical report." *arXiv preprint arXiv:2309.16609* (2023)

Summary Transformer Architectures for Language Models

Architecture	How It Works	Feature	Tasks	Examples
Encoder-Only	Processes the full input bidirectionally and creates contextual representations	Natural language understanding	Text classification, named entity recognition, information extraction, retrieval, sentence similarity	BERT, RoBERTa, ELECTRA
Encoder-Decoder	The encoder represents the input, and the decoder generates the output sequence	Sequence-to-sequence transformation and generation	Machine translation, summarization, paraphrasing, question answering	T5, mT5, BART, mBART
Decoder-Only	Predicts the next token from the preceding context and generates text autoregressively	Natural language understanding and generation through prompting	Dialogue, translation, summarization, question answering, classification, information extraction, reasoning	GPT, LLaMA, Gemma, Qwen

Part 2: Low-resource Machine Translation (Myanmar-English-Thai)

Selected Journal Paper:

Mya Ei San, Sasiporn Usanavasin, Ye Kyaw Thu, Manabu Okumura, "[A Study for Enhancing Low-resource Thai-Myanmar-English Neural Machine Translation](#)," ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP), Volume 23, Issue 4, Apr 2024, Article No.54, pp. 1:24.

Research problem

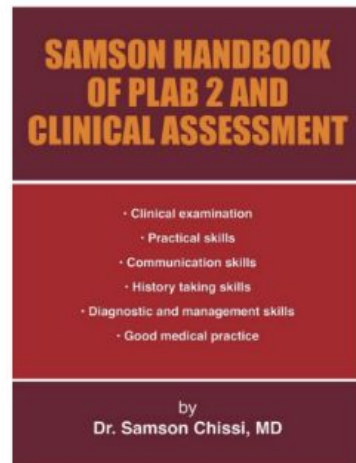
- NMT requires substantial parallel data.
- Thai and Myanmar remain comparatively low-resource.
- Data augmentation and finetuning approach had not been thoroughly evaluated across these three languages.

Main question

What approaches can improve low-resource NMT for English, Thai, and Myanmar when only limited parallel data are available?

Medical Data Collection

- Medical sentences mainly from “SAMSON HANDBOOK”
- This book includes
 - The conversation between patients and doctors
 - The names of diseases and medicine
 - The methods of treatment
- Manually translated sentences by sentences from **English to Myanmar**
- For **Thai** parallel sentences, translated from English to Thai via Google Translate =>**remains post-editing by native Thai**
- Also gathered few of English Medical sentences from <https://github.com/danhett/Doctordocotor/blob/master/story.txt>
- We collected from few of English-Thai sentences from “Dr. WATHIT CHANYAEM HANDBOOK”



Translation Setting

Three languages · Six directions

Dataset

- **ASEAN-MT parallel corpus**
- Travel-domain sentences
- Bidirectional experiments for all language pairs

Evaluation

- BLEU
- ChrF
- TER

English–Thai	English–Myanmar	Thai–Myanmar
English → Thai	English → Myanmar	Thai → Myanmar
Thai → English	Myanmar → English	Myanmar → Thai

Approaches Investigated

Baseline Models

- **Transformer**
- **EDITOR**
 - Edit-based sequence generation
 - Generates translations through insertion, deletion, and repositioning

Data Augmentation

SwitchOut

- Randomly replaces tokens in source and target sentences
- Creates noisy training examples
- Improves model robustness

CipherDAug

- Converts source sentences into ciphertext representations
- Uses original and transformed sentences as multiple inputs
- Increases data diversity while preserving structural patterns

Approaches Investigated

Multilingual Pretrained Model

mBART

- Multilingual encoder–decoder pretrained with a denoising objective
- Fine-tuned on low-resource parallel data
- Transfers multilingual knowledge to the translation task

SwitchOut Data Augmentation with Transformer

- Based on the theory of Word Dropout and RAML, SwitchOut (Wang et al., 2018) algorithm is formulated.
- The SwitchOut algorithm separately measures the similarity between the pair $(x, \hat{x})$ and the pair $(y, \hat{y})$ and finally sum them together as below equations:

$$s(\hat{x}, \hat{y}; x, y) / \tau \approx \gamma_x(\hat{x}, x) / \tau_x + \gamma_y(\hat{y}, y) / \tau_y$$

where,

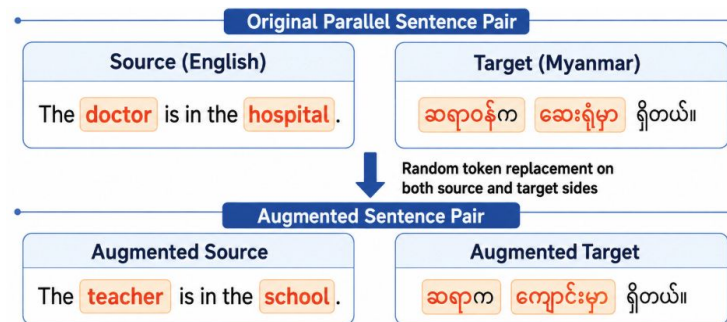
γ_x and γ_y are domain specific similarity functions

τ_x and τ_y are hyper-parameters that absorb the temperature parameter τ

x and $\hat{x}$ are original and augmented sentences of the source side

y and $\hat{y}$ are original and augmented sentences of the target side

The number of sentences to be augmented is according to the probability function : $p(\hat{n}) \propto e^{-\hat{n}/\tau}$ where, $\hat{n}$ is the number of input sentences and τ controls the deviation or weight of loss between the original and the augmented sentences



Ciphertext Based Data Augmentation with Multi-Source Transformer Training

- CipherDAug transforms each source-side character into another character using a bijective cipher such as **ROT-(k)**.
- Because the mapping is one-to-one, the ciphertext preserves important distributional properties of the original sentence.
- In our experiments, **ROT-2** was used, shifting each character by two positions.

Multi-view learning

- The original sentence and its ciphertext are treated as two different views of the same source sentence.
- They retain similar sentence length, structure, and underlying meaning.
- Both representations are used together in multi-source training.

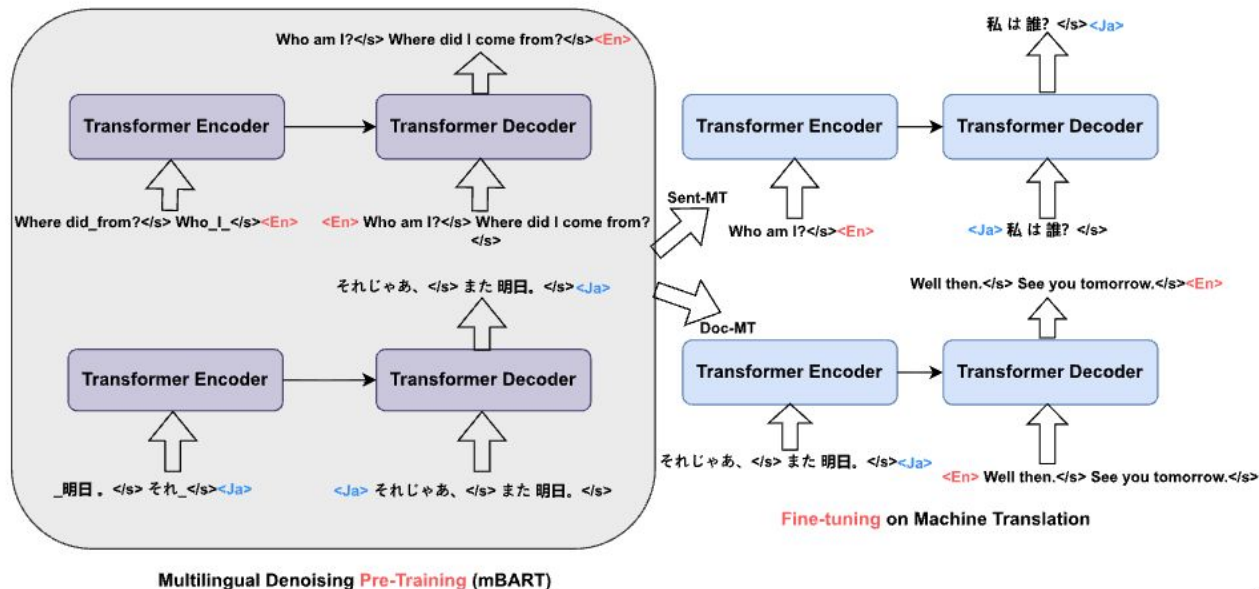
Multi-source Transformer

- Encoder 1 processes the original source sentence.
- Encoder 2 processes the ciphertext sentence.
- A shared decoder generates the target translation.
- The encoders share parameters and embedding matrices.

Plaintext (source language)	Have you been to Bangkok by train? [english] ဘန်ကောက် ကို ရထား နဲ့ သွားဖူး လား။ [myanmar] คุณ เคย ไป กรุงเทพ โดย รถ ไฟไหม? [thai]
Ciphertexts ROT-1 (source language)	Ibwf zpv cffo up Cbohlpl cz usbjo ? [english] မပဲခေခိ ခို လ အား ပဲ ဟူး ဗူး တား ။ [myanmar] ผิด แพร งาม ขฤจแธฟ ไตร ฤทฎก ฎพย ? [thai]
Ciphertexts ROT-2 (source language)	Jcxg éqw dggp vq Dcpimqm dé vtckp ? [english] ယပဲကောန် ဝို ဝ တား ပဲ အူး ဘူး သား ။ [myanmar] จ ฏ โงฤ Aค ขฤจโนก โถฤ ลคAm Aอธ ? [thai]

Finetuning mBART

- mBART pre-training follows the pattern of the BART model, which only includes English language pre-training. A Seq2Seq transformer architecture was used to implement mBART, which consists of 12 layers of encoders and 12 layers of decoders on 16 heads.
- Transfers knowledge acquired from multilingual data
- Requires less task-specific training than training entirely from scratch



Some Results

Models	English–Myanmar			Myanmar–English		
	BLEU (↑)	ChrF (↑)	TER (↓)	BLEU (↑)	ChrF (↑)	TER (↓)
Baseline Transformer	24.2	41.9	62.4	23.7	40.1	71.2
Baseline EDITOR	13.5	37.2	63.4	13.4	32.2	74.2
Transformer+SwitchOut	23.4 ^φ	40.5 ^φ	61.5	24.5^φ	38.3 ^φ	70.4
Multi-Source Transformer+CipherDAug	28.9^{★φ}	47.5^{★φ}	54.1^{★φ}	31.2^{★φ}	45.9^{★φ}	60.2^{★φ}
Fine-tuned mBART	25.7^{★φ}	46.2^{★φ}	57.2^{★φ}	23.9^φ	40.6^φ	69.6^{★φ}

A higher BLEU and ChrF score (↑) indicate better translation quality. Conversely, a lower TER score (↓) signifies improved translation quality. Scores with bold demonstrate the results that notably outperform both baseline models, at $p < 0.05$. Scores marked with a star (★) signify results that are significantly higher than those of the first baseline Transformer model, and scores marked with a varphi (φ) denote results significantly surpassing those of the second baseline EDITOR model, at $p < 0.05$.

Summary of Findings

Experimental scope

- Evaluated six translation directions among English, Thai, and Myanmar
- Compared two baselines with three enhancement approaches:
 - SwitchOut
 - Multi-Source Transformer with CipherDAug
 - Fine-tuned mBART

Main findings

- **CipherDAug achieved the best overall performance** across all translation directions in BLEU, ChrF, and TER.
- Incorporating ciphertext as an additional linguistic representation through multi-source training effectively improved low-resource NMT.
- **Fine-tuned mBART substantially outperformed the baselines** in most directions, except Myanmar → English.
- **SwitchOut improved over the EDITOR baseline** but performed similarly to the standard Transformer.

Key takeaway

Structured data augmentation and multilingual knowledge transfer are promising strategies for improving low-resource English–Thai–Myanmar NMT.

Part 3: Current work: Low-Resource Machine Translation with the LLMs

From NMT to LLM-Based Translation

Neural Machine Translation	LLM-Based Translation
Usually trained specifically for translation	Built on general-purpose pretrained language models
Relies mainly on parallel corpora	Use instructions, demonstrations, and external knowledge
Task knowledge is learned during training	Additional task knowledge can be provided in the prompt or training
Improvements require retraining or fine-tuning	Behavior is adapted through prompting, ICL, RAG, or fine-tuning

How LLMs Adapt to New Tasks

Inference-Time Adaptation

No parameter update

- **In-Context Learning (ICL)**
 - Zero-shot, one-shot, and few-shot prompting
- **Retrieval-Augmented Generation (RAG)**
 - Retrieves relevant examples or external knowledge
 - Adds them to the model context

Continued Pretraining

Parameters are updated using additional data

- Continues language-model training on domain- or language-specific text
- Helps the model acquire new terminology and distributional patterns
- Includes domain-adaptive and language-adaptive pretraining

How LLMs Adapt to New Tasks

Post-Training and Fine-Tuning

Parameters are updated using task or preference data

- **Supervised Fine-Tuning (SFT)**
 - Trains on input–output pairs
- **Instruction Fine-Tuning**
 - Uses diverse instruction–response examples
- **Parameter-Efficient Fine-Tuning (PEFT)**
 - Updates a small set of parameters, such as LoRA
- **RLHF**
 - Aligns outputs with human preferences

Key takeaway

LLMs can be adapted through prompt context, continued pretraining, or task-specific post-training.

Findings of Challenges for Myanmar Translation with LLMs

Uneven Performance Across Translation Directions

- Translation quality can differ substantially by direction.
- For example, English → Myanmar may perform better than Myanmar → English in some settings.

Fluent but Less Reliable Translation

- LLM outputs are often fluent and semantically rich.
- However, models may omit source information, add unsupported details, or alter the intended meaning.
- Fluency can make these errors difficult to detect.

Limited Control over Generation

- LLMs provide less direct control over word-level alignment and translation decisions than conventional NMT systems.
- Prompt instructions do not always guarantee consistent terminology, length, or sentence structure.

Findings of Challenges for Myanmar Translation with LLMs

Difficulty with Complex Text

- Everyday conversational sentences may be translated reasonably well.
- Performance can decrease for long, literary, domain-specific, or structurally complex sentences.

Language-Specific Challenges

- Word segmentation
- Domain Specific Terms
- Long-distance dependencies and complex clause structures

Evaluation Challenges

- Traditional metrics such as BLEU may not fully capture word by word accuracy.
- Human-centered evaluation is needed.
- More benchmark tasks and high-quality test sets are necessary for reliable comparison.

Key takeaway

LLMs can produce fluent Myanmar translations, but controlling faithfulness, handling complex structures, and evaluating quality remain major challenges.

My Current Research Direction

Improving low-resource English–Myanmar machine translation using LLMs

Key Research Questions

- How effectively can LLMs translate between English and Myanmar?
- Can LLMs translate cultural terms, domain specific terms correctly between English and Myanmar?
- Can external linguistic knowledge improve translation performance?
- How can translation errors be detected and corrected?
- How should LLM-based low-resource machine translation be evaluated?

Part 4: LLMs in Emerging domains and Lessons Learned

LLMs in emerging domains

LLMs are increasingly applied to domains where:

- Labeled data are limited
- Expert knowledge is required
- Evaluation is subjective or uncertain

Example Application Areas

- **Healthcare and Life Sciences:** clinical summarization, medical question answering, research synthesis
- **Legal and Regulatory:** contract analysis, legal-document review, compliance support
- **Finance:** financial-report analysis, market-news synthesis, audit support
- **Cybersecurity and Software Engineering:** debugging, code migration, vulnerability analysis, threat intelligence

LLMs in emerging domains

My RA Case Study: Olfactory Language

- Applying LLMs to understand and generate odor-related descriptions
- Identifying explicit and implicit odor characteristics from natural-language text
- Handling subjective labels and multiple plausible interpretations
- Evaluating outputs when reliable annotations and objective metrics are limited

Shared challenge

Like low-resource machine translation, olfactory language research involves limited supervision, ambiguous outputs, and difficult evaluation.

Industry and Academia Research: Different Perspectives

Academic Perspective

Contribution and novelty

Controlled experiments

Benchmark improvement

Reproducibility

Understanding why it works

Long-term knowledge creation

Industrial Perspective

Practical value

Real-world robustness

User impact

Scalability and maintainability

Ensuring it works consistently

Short- and medium-term delivery

Lessons from Both Environments

- Start with the clear problem, not the method.
- Establish a strong and simple baseline.
- Change one major factor at a time.
- Record the experimental logs properly.
- Analyze failures, not only average scores.
- Do not assume a more complex pipeline is better.
- Use negative results to refine the research question.
- Of course, clean and high-quality data are still essential in low-resource settings!

Personal lesson

Research progress often comes from understanding why an idea did not work as expected.

Any Question?

Thank You So Much For Your Time!